

Mobility Lens Project Report

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Application Package: com.priyabajarla.cs501.mobilitylens

Emulator: Pixel 7

Emulator API Level:API 37.1

Application Overview

Mobility Lens is a single screen Android application that I created using Kotlin and Jetpack Compose. Its purpose is to help a developer consider six differences between mobile and stationary applications. These dimensions are input and interaction, screen size and orientation, lifecycle and resource constraints, context awareness, usage patterns, and security and privacy expectations.

The application displays one dimension at a time with a short description and a practical recommendation. Previous and Next buttons allow the user to navigate through all six dimensions. The user can also enter the name of an application or feature in an outlined text field. Selecting Apply this lens displays a message that combines the entered name with the current dimension. If the field is blank, the application displays a validation message.

SDK Configuration

The application uses Minimum SDK: API 24

Target SDK: API 37

Compile SDK: API 37

The minimum SDK identifies the oldest Android version on which the application can be installed. I selected API 24 to make the application compatible with a large range of Android devices while still using modern Android development tools.

The target SDK identifies the Android version for which the application was designed and tested. It also determines which current Android behavior changes are applied to the application.

The compile SDK identifies the Android API version used to compile the source code. It controls which Android APIs are available while developing the application. The compile SDK does not determine the oldest device that can install the app because that is controlled by the minimum SDK.

Important Project Files

`MainActivity.kt` is located under `app/src/main/java/com/priyabajaria/cs501/mobilitylens`. It is the application entry point and displays the main Compose screen inside the custom Material theme.

`AndroidManifest.xml` is located under `app/src/main`. It registers `MainActivity` as the launcher activity and defines application information such as the label, icon, and theme. The application does not request permissions because its features do not require access to protected device resources.

The app level `build.gradle.kts` file is located inside the app folder. It defines the package, SDK settings, Compose configuration, and application dependencies.

The version catalog is located at `gradle/libs.versions.toml`. It keeps plugin and library versions organized in one location.

`strings.xml` is located under `app/src/main/res/values`. It stores the text displayed in the application, including the six dimensions, button labels, field labels, and validation messages.

Rotation Observation

I navigated away from the first dimension, entered an application name, and then rotated the Pixel 7 emulator. After the rotation, the application returned to the first dimension, the text field became empty, and the feedback message disappeared. This means that the selected dimension, entered text, validation state, and feedback message did not survive the rotation. The application's fixed content, theme, and layout remained available because they were recreated from the code and resources.

This happened because the screen state was stored using `remember`. It can preserve values during normal recomposition, but it does not preserve them when Android recreates the activity after a configuration change such as rotation. An Android application may need to preserve this information so that users do not lose their progress when the device orientation changes. For example, losing text that a user has already entered could be frustrating and may force the user to repeat their work.

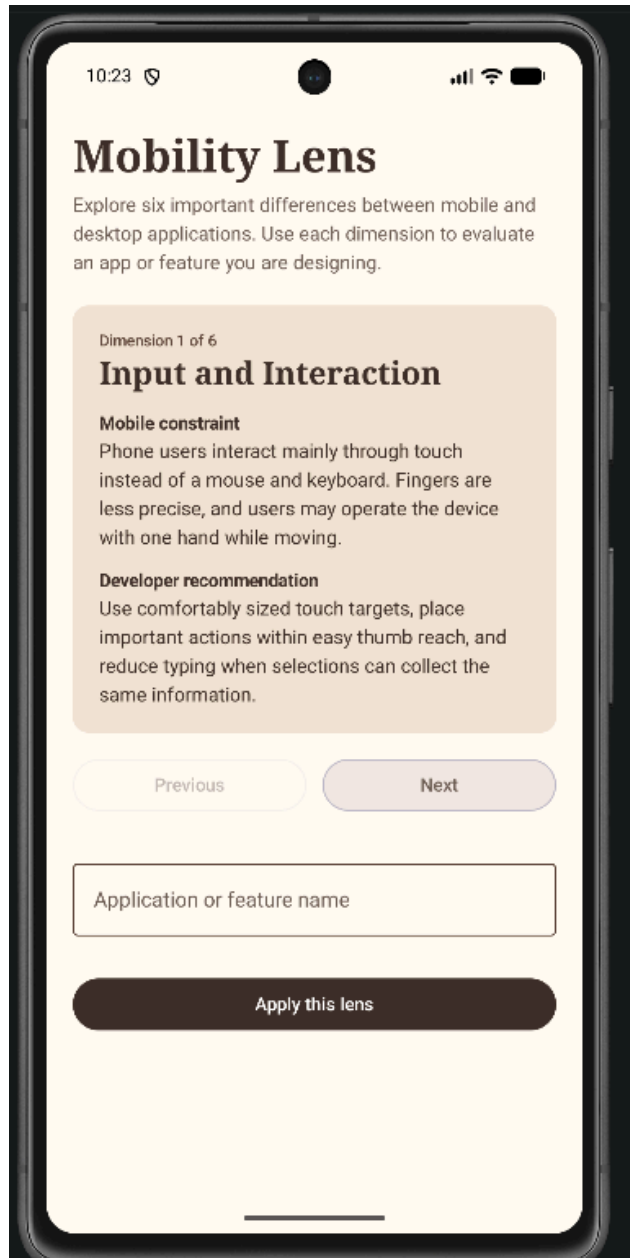
Mobile Design Decision

I used a vertically scrollable layout with large, full width controls. This design is appropriate for a phone because screen space is limited and users interact using touch rather than a mouse. The spacing and button sizes make the application easier to operate with one hand.

Application Screenshots

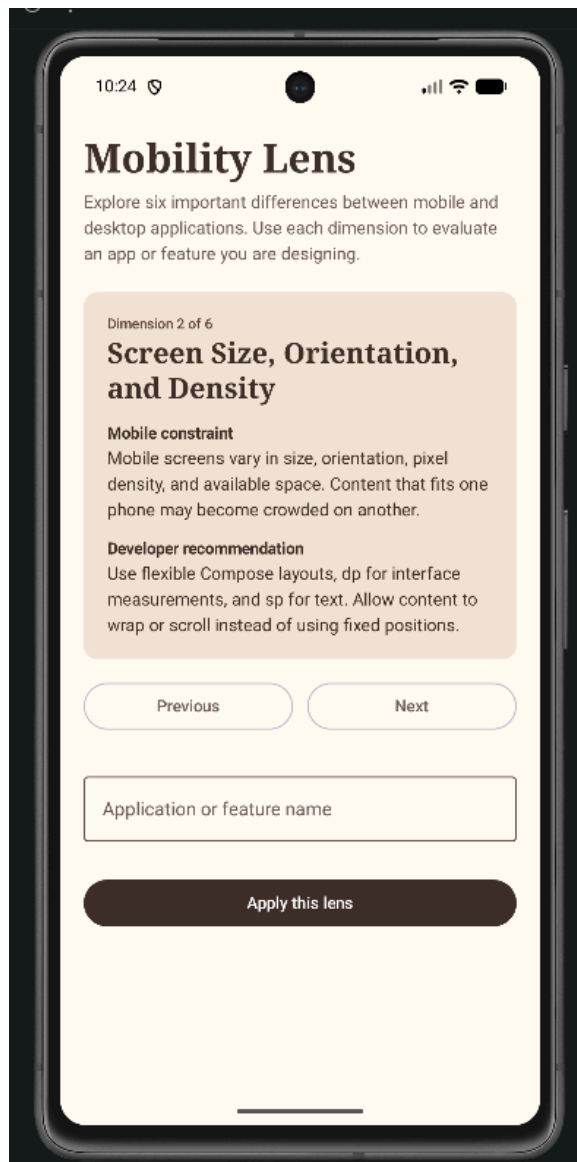
Initial Application Screen

The application running on the first dimension, Input and Interaction.



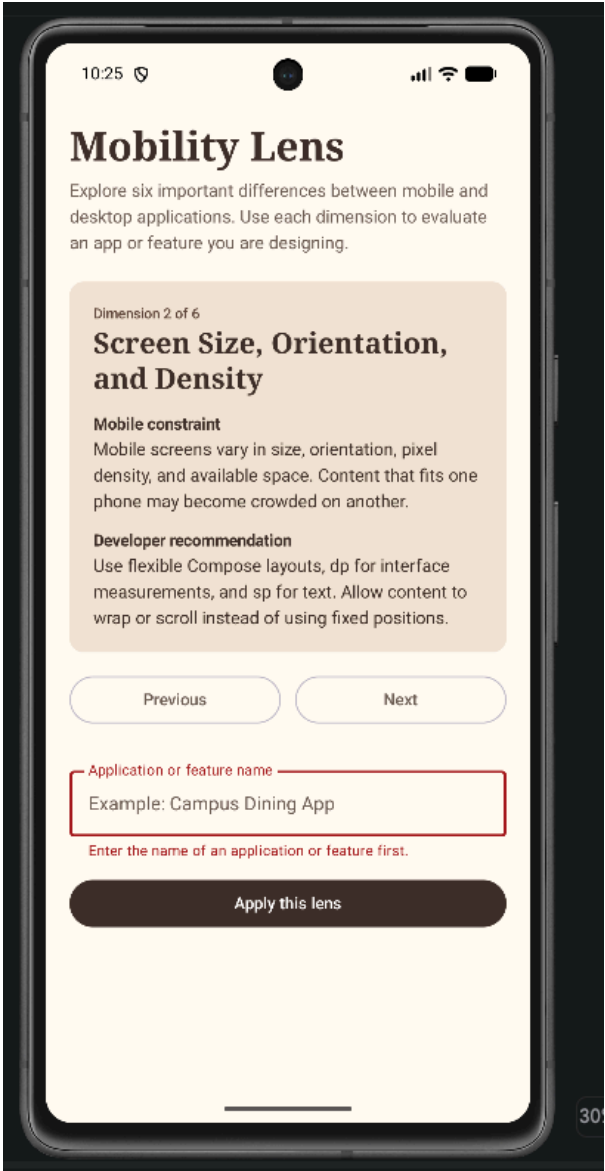
Dimension After Navigation

A different mobility dimension displayed after using the Next button.



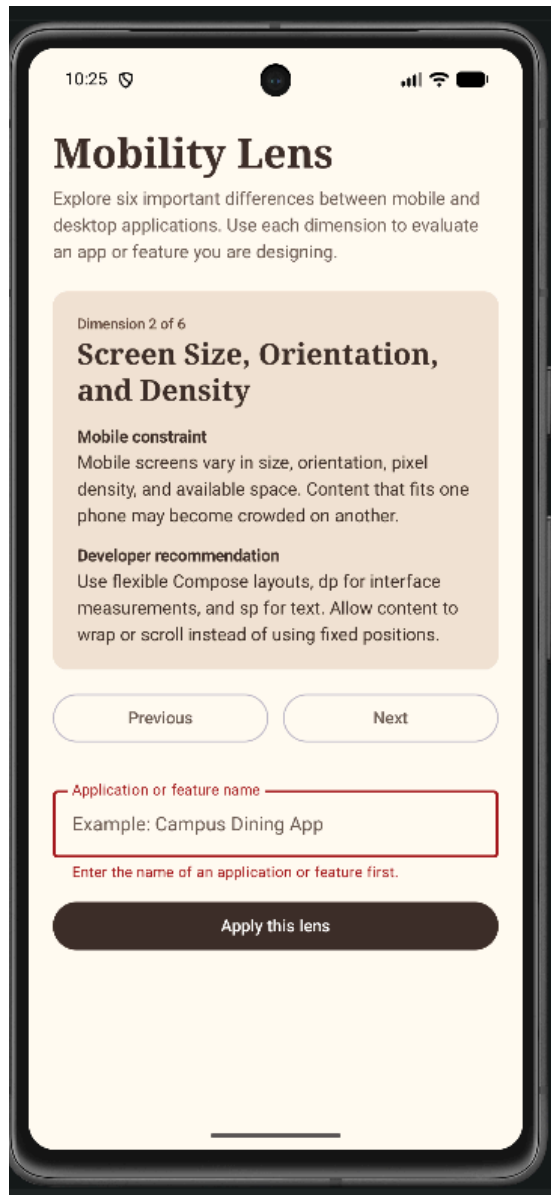
Blank Input Validation

The validation response displayed after selecting Apply this lens without entering an application or feature name.



Successful Input Response

The successful response displayed after entering a valid application name and applying the currently selected dimension.



Collaboration and Generative AI Disclosure

I used ChatGPT to help interpret the requirements, organize the project structure, review Compose code, and troubleshoot errors. I entered, tested, reviewed, and modified the code in Android Studio and verified the final application behavior.